

Umesh Kumar Komati

Software Engineering Senior Analyst

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Hyderabad

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SUMMARY

- Performance-driven Data Science professional with around 5 years of experience in building end-to-end **machine learning solutions and scalable data pipelines** using Python, SQL, and cloud technologies.
- Experienced in **data preprocessing, feature engineering, and developing predictive models** using algorithms such as Random Forest, Decision Trees, BERT, Neural Networks and LSTM.
- Proficient in designing and building **ETL pipelines using Airflow**, enabling reliable data ingestion from multiple sources for analytics and modeling.
- Strong background in **EDA, model evaluation, and optimization**, delivering actionable insights and improving business efficiency.
- Hands-on experience in healthcare and insurance, building data-driven solutions for claims prediction, coverage analysis, and workflow optimization
- Committed to delivering efficient, scalable, and insightful data-driven solutions while collaborating with cross-functional teams.
- Collaborative team player with proven ability to **translate business problems into scalable AI/ML solutions**, backed by a Master's degree in Machine Learning & AI and **20+** real-world projects.
- Known for being a quick learner with a passion for mastering new technologies and the ability to work in a dynamic environment.

KEY SKILLS

Python | Data Engineering | ETL Processes | Data Integrity | CI/CD Implementation | Cloud | Machine Learning Models |
Java | Selenium | Testing | GEN AI | Data Visualization | EDA | Data-Driven Solutions | TAMR |
Strategic Planning & Execution | Cross-Functional Collaboration | Leadership & Team Management | Scrum

TECHNICAL SKILLS

Scripting & Programming Languages: Python, Java, Oracle SQL

Data Processing and Visualization: Pandas, NumPy, Matplotlib, Seaborn, Pyplot, OBIEE, Tableau

Machine Learning & AI : Machine learning, Deep Learning, NLP, LLM, PyTorch, TesnsorFlow, Neural Networks, Data Science Libraries

Cloud Technologies & Deployment: Microsoft Azure, AWS Lambda

DevOps Tools & Version Control : Jenkins, GIT, Bitbucket, CA WA Workstation

Project Management & Tracking Tools : Maven, Zephyr, Rally, ALM, JIRA boards, Confluence

Test Automation : Selenium, Selenium WebDriver, Cucumber, TestNG

Software Development Methodologies : SDLC, Agile, Waterfall

EDUCATION

Master of Science in Machine learning and AI

Mar '22 - Jul '24

Liverpool John Moores University, United Kingdom

Liverpool, UK

• Key Projects:

- Time Series Analysis | Telecom Churn Case Study | Lexical Processing | Syntactic Processing
- Business Problem Assignment | Building Automated Data Pipelines with Airflow | Analytics using PySpark

• Libraries and models used:

- NumPy | Pandas | Seaborn | PySpark | Keras | TensorFlow | PyTorch | Scikit-learn | NLTK | Word2Vec | MLOps
- Decision Trees | Random Forest | Transfer learning | BERT | LSTM | CNN | RNN | LLM | Clustering | Naive Bayes | SVM | GPT

Post Graduate Diploma in Machine learning and AI

Mar '22 - Jun '23

IIIT Bangalore

Bengaluru, IN

• Course Modules:

- Data Analysis using SQL | Introduction to Python | Machine Learning

Bachelor of Engineering in Electronics and Communication Engineering

Jun '18 - May '21

MVSR Engineering College, Hyderabad - Osmania University

Hyderabad, IN

Percentage - 81%

Secondary School Certificate

Jun '14 - Apr '15

Bhashyam high school, Hyderabad

Hyderabad, IN

GPA - 9.3/10

PROFESSIONAL EXPERIENCE

Software Engineering Senior Analyst

Nov '25 - Present

Evernorth Health Services

Hyderabad, IN

- Developed and deployed a GenAI-based chatbot leveraging LLMs to enable business users to query spend data using natural language
- Built interactive dashboards and KPI visualizations to provide real-time insights and enhance business decision-making
- Championed **cross-functional team initiatives** to pinpoint artificial intelligence & machine learning use cases, gathering technical requirements & ensuring alignment with business needs, while deploying models for real-time analytics solutions across multiple departments
- Designed and developed **Data Science POCs** to evaluate feasibility of AI/ML use cases, accelerating decision-making and solution adoption.
- Collaborated with the **Opkey automation team** to develop and maintain automation scripts, improving test coverage and reducing manual effort.
- Worked on **data preprocessing and exploratory analysis**, ensuring high-quality datasets for modeling and analytics use cases.
- Partnered with cross-functional teams to translate business problems into **data-driven solutions**, contributing to end-to-end implementation from POC to deployment
- **Led** and **mentored** junior team members, guiding them to enhance skills and ensure high-quality **deliverables**

Software Engineering Analyst

Aug '21 - Oct '25

Accenture Solution Pvt Ltd

Hyderabad, IN

- Enhanced strategic planning & architectural design processes, implementing scalable solutions using **Python** to boost operational efficiency by 20%
- **Collected and cleaned** data from multiple sources, ensuring high data quality and integrity across datasets.
- Built and optimized machine learning models and applied advanced analytics techniques to derive actionable insights from large datasets
- Leveraged data-driven approaches including EDA, feature engineering and model tuning to improve business outcomes and support decision making
- **Performed** feature engineering to prepare data for machine learning models and downstream applications.
- Optimized analytic systems & predictive models using machine learning techniques, enhancing accuracy by 15%
- Collaborated across teams to resolve dependencies and undertook initiatives to increase the team contribution margin by 25%.
- **Initiated** and supported innovation activities by introducing new ideas and working on their execution for process improvement.

PROJECTS

AI-Powered Spend Analytics Chatbot & Dashboard - LLM and NLP

Python, LLM, NLP, Deep Learning, SQL, React, Tableau, AWS

An AI-powered chatbot using LLMs to enable natural language querying of spend data and built an interactive dashboard with dynamic KPIs and visualizations to deliver real-time insights and support data-driven decision-making

- Designed and developed an AI-powered chatbot using Large Language Models to enable business users to query spend data using natural language
- Trained and fine-tuned LLM models on diverse spend datasets to improve contextual understanding and response accuracy
- Built a data pipeline to ingest, preprocess, and structure spend data from multiple sources for model consumption
- Implemented prompt engineering techniques to optimize chatbot responses for business-specific queries
- Integrated the chatbot with backend data systems to fetch real-time insights and ensure accurate responses
- Developed an interactive web application to display key KPIs and graphical dashboards for business monitoring
- Designed dynamic visualizations and real-time KPI trackers to provide actionable insights on spend trends
- Enabled natural language to SQL/query translation for seamless interaction between users and data systems
- Collaborated with stakeholders to identify key business metrics and tailor chatbot responses accordingly
- Improved decision-making efficiency by reducing dependency on manual reporting and enabling self-service analytics

TAMR - Data analytics and Machine learning

Python, SQL, Airflow, Machine learning, deep learning, NLP, TAMR tool, Opkey

A healthcare application where collaborated with cross-functional teams to gather requirements, developing ML models, and creating end-to-end data pipelines to process and integrate model outputs, ensuring efficient data flow and improving overall system performance

- Designed and implemented an **end-to-end data pipeline** integrating outputs from machine learning models into downstream ETL systems using Python and Airflow DAGs.
- Developed and scheduled **Airflow pipelines** to automate data ingestion, transformation, and workflow orchestration.
- Created **data visualizations in Tableau** to present insights and support data-driven decision-making.
- Collaborated with business team to understand common data model requirements and built a scalable workforce analytics data model for efficient data processing and analysis.
- Utilized **TAMR tool** to create data clusters and build models for enhanced data organization and insights.
- Developed and maintained automation scripts using Opkey, ensuring reliable test execution and supporting continuous testing efforts.
- Built **machine learning models** using AWS Lambda Lab and Python, implementing algorithms like Decision Trees, Random Forests, BERT, and LSTM.
- Experienced in creating Airflow Dags for automated monthly ETL runs by developing data pipelines.
- Created **data visualizations** using Matplotlib to communicate insights effectively and support data-driven decision-making

FAH claims - Data analytics and Machine learning

Python, AWS Lambda, SQL, S3, NumPy, Pandas

An insurance based application to identify coverage & claims data by using derivation rules having numerous null values. After business discussion, built an ML model to trace down the abnormalities and produce proper coverages & claims data on monthly runs

- Performed Exploratory data analysis to get insights from the datasets.
- Created workflows for cleaning and preprocessing the datasets.
- Performed feature engineering to create new features that add value to the final solution.
- Created the machine learning model that predicts the final derived claim value and coverage type of customer
- Created end-to-end process from data exploration to fine-tuning the model implemented using various machine learning techniques.

Healthcare finance - ETL and automation

Selenium, Java, Cucumber, ETL, SQL, Jenkins, Confluence, Jira, Scrum, zephyr

An ETL dependency where need to develop an ETL pipeline from scratch and populate the transformed data to UI and test the whole UI application designed by dev team using automation like selenium java

- Proficient in designing and implementing robust automation frameworks, leveraging expertise in Selenium, Core Java, and TestNG to ensure efficient test execution and software quality enhancement.
- Proven record of identifying automation opportunities, ensuring efficient test execution, and generating detailed reports.
- Developed and executed ETL test cases using SQL and automated the runs using Jenkins jobs to generate the evidence
- Contributing to Agile ceremonies, effectively handling US clients.

ACCOMPLISHMENTS/ REWARDS

- Won Pinnacle award for contribution in work and getting project funding from clients.
- Won Applause Awards 5 times for outstanding performance in the span of 2 years in Accenture.
- Won Friend in need award for helping other core teams during peak work time by learning their work within no time and achieved great appreciations from clients.
- Won many awards in chess and cricket competitions held within the company.

CERTIFICATIONS

- **AZ-900: Microsoft Azure Fundamentals**
- **Oracle Certified foundation associate**

Additional Competencies:

- Possess Good knowledge in ETL, SQL, SQL Server and Data warehouse concepts.
- Good at documenting, test plan, test results, test strategy, test summary report and test cases.
- Driven deadline meetups, Scrum, Standup & Ad-hoc meetings.
- **Adapted** to new technologies and continuously **upgraded** skills to stay current with industry advancements
- Good in technical, logical skills and strong analytical skills.
- Ability to manage team and provide knowledge transfer to the peers.
- Effective communication across multiple teams.